PARTS 4-6

Performance comparison

Milliseconds for all timings.

Matrix size	CPU	Naïve CUDA	Tiled CUDA	cuBLAS	cuBLAS + pinned memory
512 × 512	220.179	2.264	2.169	1.947	0.605
1024 × 1024	3,449.639	9.249	7.866	6.012	2.394
2048 × 2048	63,882.329	59.962	40.732	22.987	10.379

Speedup over CPU

Speedup is calculated using the total GPU time including transfers, rather than kernel time alone.

Matrix size	Naïve CUDA	Tiled CUDA	cuBLAS	cuBLAS + pinned memory
512 × 512	97.3×	101.5×	113.1×	364.0×
1024 × 1024	373.0×	438.5×	573.9×	1,440.4×
2048 × 2048	1,065.4×	1,568.2×	2,779.4×	6,155.9×